

Geometry
Unit 10
Lesson 5 Practice

Name Answer Key
Date
Period

1. Circle C has points X, N , and P on the circle with $m\angle NCX = 47.1^\circ$. If $m\widehat{NP} = 116.9^\circ$ and $\widehat{NP}$ does not intersect $\widehat{NX}$, what is $m\angle PCX$?

$$m\angle PCX = 196^\circ$$

2. Circle S has radii $\overline{ST}$ and $\overline{SR}$. Find the value of x if $m\angle RST = 135^\circ$ and $m\widehat{RT} = (6x + 9)^\circ$.

$$6x + 9 = 135$$

$$6x = 126$$

$$x = 21$$

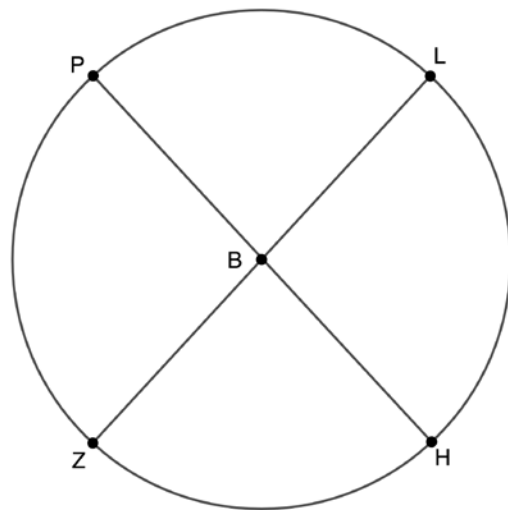
For questions 3-4, Circle B has diameters $\overline{PH}$ and $\overline{LZ}$ that both intersect at point B .

3. Given $m\widehat{HZ} = 72^\circ$, find the $m\widehat{PL}$.

$$m\widehat{PL} = 72^\circ$$

4. Given $m\widehat{HL} = 108^\circ$, find the $m\widehat{ZP}$.

$$m\widehat{ZP} = 108^\circ$$



For questions 5-6, consider the following context.

Lorenzo is creating a folding fan. He needs to determine the measure of the arc of the fan before he can cut the paper. He uses the expression $(6x + 110)^\circ$ as the measure of the central angle, $\angle ARQ$. For the $m\widehat{QA}$, he uses the expression $(24x + 38)^\circ$.

5. What is the value of x ?

$$6x + 110 = 24x + 38$$

$$72 = 18x$$

$$x = 4$$

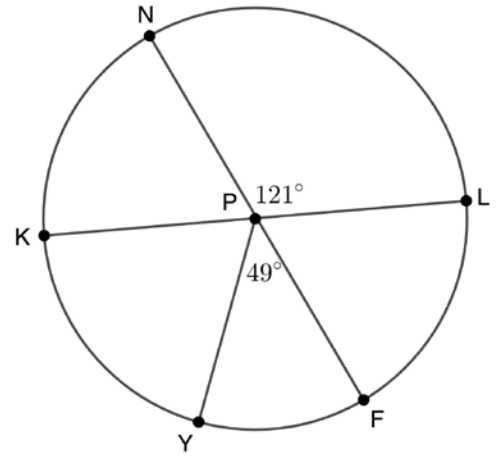
6. What is the degree measure of the arc of the fan, $\widehat{QA}$?

$$134^\circ$$

7. In Circle P , $m\angle NPL = 121^\circ$ and $m\angle FPY = 49^\circ$. Calculate the $m\widehat{NLY}$, given that P is the center and $\overline{NF}$ and $\overline{KL}$ are diameters.

$$\overline{LF} = 180 - 121 = 59^\circ$$

$$m\widehat{NLY} = 121 + 59 + 49 = 229^\circ$$



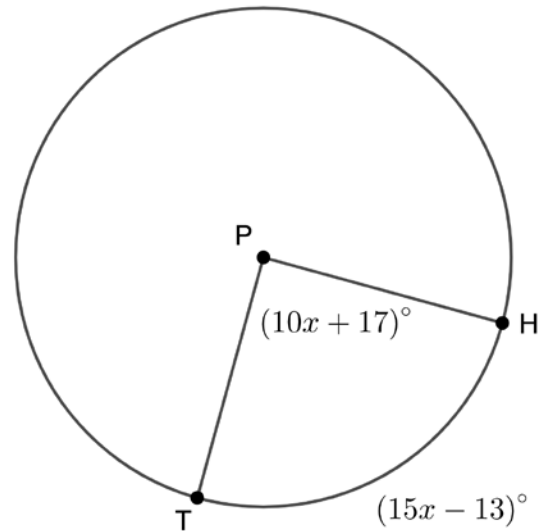
8. Circle S has points K, D , and Y on the circle.
 $m\angle YSD = 71^\circ$, $m\widehat{KY} = 132^\circ$, and $\widehat{KY}$ does not intersect $\widehat{YD}$.
 Find $m\angle KSD$.

$$360 - 71 - 132 = 157$$

$$m\angle KSD = 157^\circ$$

For questions 9-10, consider the following context.

A horse is tied to a post at point P and is grazing in a circular pattern. She decides to walk from her position at point H to the trough of water at point T along the circumference of the circular area she can roam. The measure of the central angle, $\angle TPH$, is $(10x + 17)^\circ$, and the measure of the arc that the horse walked, $\widehat{TH}$, is $(15x - 13)^\circ$.



9. Find the value of x .

$$10x + 17 = 15x - 13$$

$$30 = 5x$$

$$x = 6$$

10. Find the measure of the arc that the horse walked, $\widehat{TH}$.

$$\widehat{TH} = 15(6) - 13 = 77^\circ$$